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Q3. Briefly explain the reasons behind wage and price stickiness.

Stickiness is a theoretical market condition wherein some nominal price resists change i.e when the country doesn't operate at right potential, rather the economy operates above or below potential output. While it often applies to wages, stickiness may also often be used in reference to prices within a market, which is also often called price stickiness. The entry of wage-stickiness into one area or industry sector will often bring about stickiness into other areas due to competition for jobs and companies' efforts to keep wages competitive. When stickiness enters the market a change in one direction will be favored over a change in the other. Since wages are held to be sticky-down, wage movements will trend in an upward direction more often than downward, leading to an average trend of upward movement in wages. Stickiness is also thought to have some other relatively wide-sweeping effects on the global economy.

When a person signs a wage contract, nominal wages are fixed till the contract exists. The life of such contracts vary from months to years depending on the profession of the person. Now when such contracts are signed, the wages remain fixed for employees and employers no matter the changes in the economy of the country. So that when that person serves his term at office, his wage does not fluctuate regularly with changes in supply or demand. This way that his wage is stuck, over a formal or an informal agreement for a particular amount of time frame, is known as a sticky price.

The reason why workers and firms accept long-term nominal wage contracts is, negotiating a contract is a costly process. Both parties must keep themselves adequately informed about market conditions. Where unions are involved, wage negotiations raise the possibility of a labor strike, an eventuality that firms may prepare for by accumulating additional inventories, also a costly process. Even when unions are not involved, time and energy spent discussing wages takes away from time and energy spent producing goods and services. In addition, workers may simply prefer knowing that their nominal wage will be fixed for some period of time.

Another reason might be that the firm is concerned that while the aggregate price level is rising, the prices for the goods and services it sells might not be moving at the same rate. Also, cost-of-living or other contingencies add complexity to contracts that both sides may want to avoid. Even markets where workers are not employed under explicit contracts seem to behave as if such contracts existed. In these cases, wage stickiness may stem from a desire to avoid the same uncertainty and adjustment costs that explicit contracts avert.

As wages are a large component of overall cost of doing business, wage stickiness itself leads to price stickiness. In cases, with firms whose nominal wages are stable, they will take a time period before changing or increasing their prices. In those very days, they'll analyze the sales, which might be rising or falling, try to assess likely action by customers if prices are increased etc. Meanwhile firms will adjust output, product prices, employment etc. The adjustment costs are assumed to be associated with changing prices, this is one of the reasons for price stickiness.

Another reason is that firms may have explicit long-term contracts to sell their products to other firms at specified prices eg, electric utilities etc who buy goods under long term contracts.

These are the reasons behind wage and price stickiness.

Q4. Derive the Aggregate Supply curve from Phillips curve.

Ans) The aggregate supply curve can be derived from the Phillips curve in a few steps.
They are as follows:

1. Finding the correlation between the output and employment. Okun's Law.
Okun's law is an empirical relationship that measures the correlation between the deviation of the unemployment rate from its natural rate and the deviation of output growth from its potential, in the US economy. Unemployment and output are closely related here. It states that, whenever unemployment extras by 1%, that costs 2% of GDP Output(Y), which is assumed to be proportional to employment.

Thus the production function is $Y=aN$

[a= input coefficient or labour productivity]

[Y= Output produced]

[Y/N = amount of output produced per unit of labour]

[N= amount of labour employed]

2. Labour cost is one of the major parts of the total cost. The price of output will be fixed by the firm based on which covers its cost. They may want to charge more than cost but the competition from other existing firms and new firms entering the industry, prevents them from hiking prices. Let's assume the firms keep their price on the labour cost. And assume each unit labour (L), produces 'a' unit output.

Per unit labour cost of production = W/a

Per unit cost of production = per unit labour cost of production + some margin

Therefore, $P = w/a + z*w/a$ (where, Z = other cost + profit margin)

:: Firm fixed price as a mark-up', (Z), on labour cost

$$P = (1+Z)*W/a \quad \dots (2)$$

If $a = 3$, that is, each unit of L produces 3 units of output.

If each labour input is paid 5

Then labour cost per unit of output = $w/a = 15/3 = 5$

3. Phillips curve relationship between employment and wages:

$$W_{t+1} = w_t[1 + e(N-N^*)/N^*] \quad \dots (3)$$

Derivation of AS curve :

$Y/a = N$ from (1)

$P = \{(1+z)/a\} * W \quad \dots \text{from (2)}$

(where $p_{t+1} = \{(1+z)/a\} * W_t$; $P_t = \{(1+z)/a\} w_t$)

P_{t+1} : price this year

P_t : price previous year

$$W_{t+1} = w_t [1 + e(\{N - N^*\}/N^*)] \quad \dots \text{ from (3)}$$

Putting the value of w , from (3) in equation (2)

$$P_{t+1} = P_t [1 + e(\{N - N^*\}/N^*)] \quad \dots (5)$$

$$\text{As } Y/a = N \text{ or } N = Y/a \text{ or } n^* = Y^*/a$$

Put the values in equation (5)

$$P_{t+1} = P_t [1 + e(\{(Y/a) - (Y^*/a)\}/(Y^*/a))]$$

$$P_{t+1} = P_t [1 + e((Y - Y^*)/Y^*)]$$

By defining, $\lambda = e/Y^*$, we get the AS curve

$$P_{t+1} = P_t [1 + \lambda(Y - Y^*)] \quad \dots (7)$$

.. AS curve shows the relationship between output level and prices Y, P , → At point B (a) At $Y^* \rightarrow P_{t+1} = P_t \rightarrow$ at point B

$Y = Y^* \rightarrow$ Full employment level of output

.. Prices today P_{t+1} = Prices last year

(b) If $Y > Y^* \rightarrow$ overemployment → At point C ($Y_1 > Y^*$)

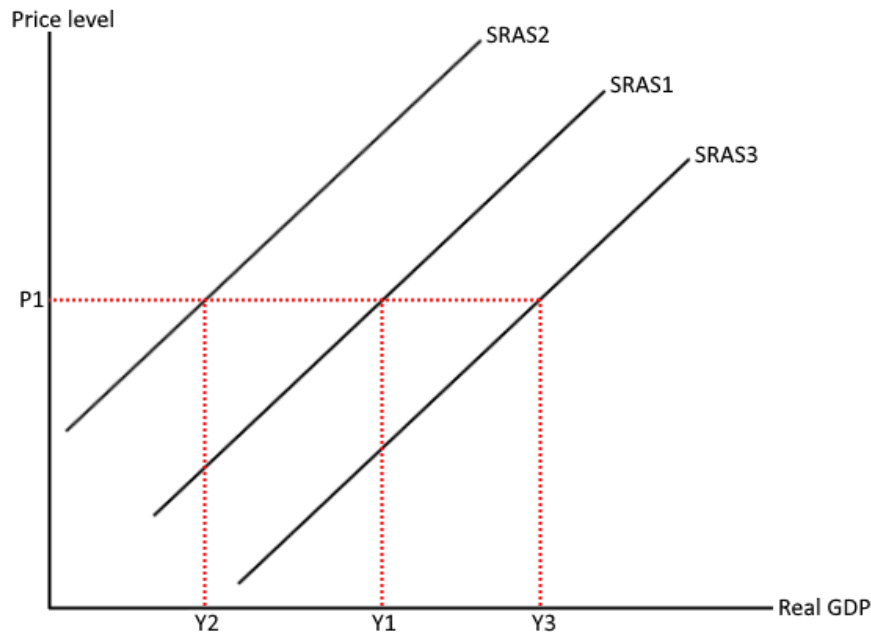
Then Prices today > Prices last year

(c) If $Y < Y^* \rightarrow$ underemployment → At point A ($Y_2 < Y^*$)

Prices today < Prices last year

Join points A, B and C and get the AS curve.

4. Put the three components together to derive an upward sloping AS curve:



Properties of Aggregate Supply Curve:

1. It is an upward sloping curve and shifts over time. So the actual output differs from full employment output level.

a. If actual output less than full employment level:

The positive AS curve is derived from the Phillips curve on the basis of 3 assumptions: (i) Mark up is fixed at Z (ii) Output is proportional to employment (iii) Wages are less than fully flexible (that is, wages are rigid/sticky).

b. If output is above full employment level :

$Y_1 > Y^* \rightarrow$ AS curve in the next period will shift upward to AS1. Actual output in current year $>$ full employment output.

Wages will increase because increase in output will lead to an increase in the demand for labour which in turn will lead to an increase in wages. As wages are the cost of production therefore increase in wages will lead to an increase in cost of production. This increase in cost will be passed on to increased prices and thus price will increase.

2. If AS is flat $\rightarrow$ Impact of output and employment on current wage is less, that is, wages respond less to unemployment.

3. Position of the AS curve depends on last year's price.

Q 5 Using the concept of expectations augmented Phillips curve establishes the wage-employment relationship. What is meant by stagflation?

Ans: The Phillips curve is an economic concept and was developed by A. W. Phillips. It stated that inflation and unemployment have a stable and inverse relationship. The theory claims that with economic growth comes inflation, which in turn should lead to more jobs and less unemployment. However, the original concept has been somewhat disproven empirically due to the occurrence of stagflation in the 1970s, when there were high levels of both inflation and unemployment.

The Phillips curve is a single-equation economic model, named after William Phillips, describing an inverse relationship between rates of unemployment and corresponding rates of rises in wages that result within an economy. Stated simply, decreased unemployment, (i.e., increased levels of employment) in an economy will correlate with higher rates of wage rises. Phillips did not himself state there was any relationship between employment and inflation; this notion was a trivial deduction from his statistical findings. Samuelson and Solow made the connection explicit and subsequently Milton Friedman and Edmund Phelps put the theoretical structure in place. In so doing, Friedman was to successfully predict the imminent collapse of Phillips' a-theoretic correlation.

While there is a short run tradeoff between unemployment and inflation, it has not been observed in the long run. In 1967 and 1968, Friedman and Phelps asserted that the Phillips curve was only applicable in the short-run and that, in the long-run, inflationary policies would not decrease unemployment. Friedman then correctly predicted that in the 1973–75 recession, both inflation and unemployment would increase. The long-run Phillips curve is now seen as a vertical line at the natural rate of unemployment, where the rate of inflation has no effect on unemployment. In the 2010s, the slope of the Phillips curve appears to have declined and there has been controversy over the usefulness of the Phillips curve in predicting inflation. Nonetheless, the Phillips curve remains the primary framework for understanding and forecasting inflation used in central banks.

Stagflation refers to an economy that is experiencing a simultaneous increase in inflation and stagnation of economic output. Stagflation is a great example of how real-world economic data can sometimes run roughshod over widely accepted economic theories and policy prescriptions. It is characterized by slow economic growth and relatively high unemployment—or economic stagnation—which is at the same time accompanied by rising prices (i.e. inflation). Stagflation occurs when money supply is expanding while supply is being constrained. For example, if a government prints currency, which would increase the money supply and create inflation, while raising taxes, which would slow economic growth, then the end result would be stagflation. Also defined as a period of inflation combined with a decline in gross domestic product (GDP).

Q6 :When is monetary policy said to be accommodating? With the help of an IS-LM framework, explain the working of an investment subsidy.

Ans:

Monetary policies that are considered accommodative include lowering the federal funds rate. They are meant to make money less expensive to borrow and encourage more spending. Accommodative monetary policy, also known as loose credit or easy monetary policy, occurs when a central bank (such as the Federal Reserve) attempts to expand the overall money supply to boost the economy when growth is

slowing (as measured by GDP). The policy is implemented to allow the money supply to rise in line with national income and the demand for money. When the economy slows down, the Federal Reserve can implement an accommodative monetary policy to stimulate the economy. It does this by running a succession of decreases in the Federal funds rate, making the cost of borrowing cheaper. The Fed can also allow the money supply to increase or increase the money supply via quantitative easing (QE). Accommodative monetary policy is triggered to encourage more spending from consumers and businesses by making money less expensive to borrow through the lowering of short-term interest rates. When money is easily accessible through banks, the money supply in the economy increases. This leads to increased spending. When businesses can easily borrow money, they have more funds to expand operations and hire more workers, which means that the unemployment rate will decrease. On the other hand, people and businesses tend to save less when the economy is stimulated due to the low savings interest rates offered by banks. Instead, any additional funds are invested in the stock market, pushing up stock prices.

Suppose IS-LM -BP model under flexible exchange rate and perfect capital mobility is equilibrium where IS, LM and BP curve intersects and defines Y_0 level of income

1. Restrictive monetary policy

The decrease in money supply as a result of restrictive monetary policy shifts LM curves to left. It increases demand for domestic currency because there is infinite capital inflows to the nation from foreigners. It causes the exchange rate to fall. It appreciates domestic currency. It decreases net exports of the nation and shifts the IS curve to left. It reduces income from Y_0 to Y_1 .

2. Expected depreciation

Expected depreciation will make depreciation of domestic currency. It increases the exchange rate. Thereby exports will increase and imports will decrease. It raises net exports and thereby IS curve shift rightward. It increases income level to Y_1 .

3. Expansionary fiscal policy

Expansionary fiscal policy shift IS curve to right. It will increase the real rate of interest. As a result, It increases demand for domestic currency because there is infinite capital inflows to the nation from foreigners. It causes the exchange rate to fall. It appreciates domestic currency. It decreases net exports of the nation and shifts the IS curve to left. It reduces income from Y_0 to Y_1